Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: Chan Chun Yuen Hayden

Student ID: 22087015d

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf 2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial 0/0/0 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129 2

Does R2 show up as an OSPF neighbor on R1? Yes 2

Does R2 show up as an OSPF neighbor on R3? No 2

What does this information tell you?

When only R2's S0/0/0 is re-activated, R2 can only form adjacency with R1. R3 still learns about the 192.168.2.0/24 network via R1 (through LSA propagation), but R3 cannot form a direct neighbor relationship with R2. This demonstrates that passive interfaces block OSPF Hello packets and prevent neighbor adjacencies, while routing information can still be propagated indirectly through other active links. 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1 1

R2(config-router)# no passive-interface s0/0/1 1

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 — R3's outgoing interface S0/0/1 cost (64) + R2's G0/0 cost (1) = **65.** OSPF cost = Reference Bandwidth (100,000 Kbps) / Interface Bandwidth. Serial at default 1544 Kbps = 64; GigabitEthernet = 1. 2

Is R2 listed as an OSPF neighbor to R3? Yes 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default reference bandwidth is 100 Mbps, which means any interface running at 100 Mbps or faster (FastEthernet, GigabitEthernet, 10G Ethernet) all get a cost of 1, making OSPF unable to differentiate between them. By increasing the reference bandwidth (e.g., to 1000 Mbps or 10000 Mbps), OSPF can assign distinct, meaningful costs to higher-speed interfaces, ensuring optimal path selection in modern networks.

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Step 2: Change the bandwidth for an interface.

g. Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

After applying bandwidth 128 on serial interfaces, OSPF recalculates cost as: 100,000 Kbps / 128 Kbps = **781** per serial interface.

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- **192.168.3.0/24 from R1:** R1 S0/0/1 (cost 781) + R3 G0/0 (cost 1) = **782**
- **192.168.23.0/30 from R1:** Two equal-cost paths — R1 S0/0/0 (781) + R2 S0/0/1 (781) = **1562**, and R1 S0/0/1 (781) + R3 S0/0/1 (781) = **1562** — both appear in the routing table.

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why? 2

1562 — With all serial interfaces now set to bandwidth 128 Kbps, each has a cost of 781. The path from R1 to 192.168.23.0/30 traverses two serial links regardless of direction: 781 + 781 = **1562**. Both paths (via R2 and via R3) are equal-cost, so both appear.

Step 3: Change the route cost.

c. Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Applying ip ospf cost 1565 on R1's S0/0/1 makes the direct path to R3 prohibitively expensive. OSPF now compares:

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- **Direct via S0/0/1 → R3:** 1565 (S0/0/1) + 1 (R3 G0/0) = **1566**
- **Indirect via S0/0/0 → R2 → R3:** 781 (R1 S0/0/0) + 781 (R2 S0/0/1) + 1 (R3 G0/0) = **1563**

Since 1563 < 1566, OSPF selects the lower-cost path through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   Without explicit control, the router ID is derived from the highest active IP address, which can change if interfaces go up/down, causing OSPF instability and neighbor relationship resets. Explicitly assigning router IDs (via loopback or router-id command) ensures each router has a stable, unique, and predictable identity in the OSPF domain, making network management, troubleshooting, and adjacency stability more reliable.

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2. Why is the DR/BDR election process not a concern in this lab?

   DR/BDR elections only occur on multi-access network types (e.g., Ethernet broadcast segments with multiple routers). In this lab, all inter-router connections use point-to-point serial links, which are in POINT_TO_POINT state as confirmed by show ip ospf interface. On point-to-point links, routers form full adjacencies directly without electing a DR or BDR.

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3. Why would you want to set an OSPF interface to passive?

   Setting an interface as passive stops it from sending or receiving OSPF Hello packets, preventing unwanted neighbor adjacencies from forming on that interface. This is typically applied to LAN interfaces connected only to end hosts (like G0/0 connected to PCs) where no OSPF routers exist. Benefits include reduced bandwidth usage, improved security (rogue routers cannot form adjacencies), and faster convergence since fewer Hello timers need tracking — while the network prefix is still advertised to OSPF neighbors on active interfaces.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   OSPF's **area-based design** provides hierarchy and scalability. By dividing large networks into multiple areas all connected to a backbone **Area 0**, OSPF limits LSA flooding and SPF recalculation to within each individual area. Routers only maintain a full LSDB for their own area, drastically reducing memory usage and CPU overhead. Inter-area routing is summarized at Area Border Routers (ABRs), allowing OSPF to scale to enterprise and service provider networks with thousands of routers.

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   The wildcard mask 0.255.255.255 matches any address beginning with 10, covering the entire 10.0.0.0/8 range.

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6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   Typical AD values are:

   - EIGRP (internal): 90
   - OSPF: 110
   - RIP: 120

Because EIGRP has the lowest AD (90), the EIGRP route is selected, even though its metric (4,039,043) is larger than the others.

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